

Hide Artifacts

Sub-techniques (14)

Adversaries may attempt to hide artifacts associated with their behaviors to evade detection. Operating systems may have features to hide various artifacts, such as important system files and administrative task execution, to avoid disrupting user work environments and prevent users from changing files or features on the system. Adversaries may abuse these features to hide artifacts such as files, directories, user accounts, or other system activity to evade detection.^{[1][2][3]}

Adversaries may also attempt to hide artifacts associated with malicious behavior by creating computing regions that are isolated from common security instrumentation, such as through the use of virtualization technology.^[4]

ID: T1564

Sub-techniques: [T1564.001](#), [T1564.002](#), [T1564.003](#), [T1564.004](#), [T1564.005](#), [T1564.006](#), [T1564.007](#), [T1564.008](#), [T1564.009](#), [T1564.010](#), [T1564.011](#), [T1564.012](#), [T1564.013](#), [T1564.014](#)

- Tactic: [Stealth](#)
- Platforms: ESXi, Linux, Office Suite, Windows, macOS
- Version: 2.0
- Created: 26 February 2020
- Last Modified: 12 May 2026

[Version Permalink](#)

Procedure Examples

ID	Name	Description
S0482	Bundlore	Bundlore uses the <code>mktemp</code> utility to make unique file and directory names for payloads, such as <code>TMP_DIR=`mktemp -d -t x`</code> . ^[5]
S1066	DarkTortilla	DarkTortilla has used <code>%HiddenReg%</code> and <code>%HiddenKey%</code> as part of its persistence via the Windows registry. ^[6]
S9025	NOOPLDR	NOOPLDR can hide services used to aid execution. ^[7]
S0402	OSX/Shlayer	OSX/Shlayer has used the <code>mktemp</code> utility to make random and unique filenames for payloads, such as <code>export tmpDir="\$(mktemp -d /tmp/XXXXXXXXXXXX)"</code> or <code>mktemp -t Installer</code> . ^{[8][5][9]}
S0332	Remcos	Remcos can modify file attributes to hide the file. ^[10]
S1011	Tarrask	Tarrask is able to create "hidden" scheduled tasks by deleting the Security Descriptor (<code>SD</code>) registry value. ^[11]
S0670	WarzoneRAT	WarzoneRAT can masquerade the Process Environment Block on a compromised host to hide its attempts to elevate privileges through <code>IFileOperation</code> . ^[12]

Mitigations

ID	Mitigation	Description
M1049	Antivirus/Antimalware	Review and audit file/folder exclusions, and limit scope of exclusions to only what is required where possible. ^[13]
M1013	Application Developer Guidance	Application developers should consider limiting the requirements for custom or otherwise difficult to manage file/folder exclusions. Where possible, install applications to trusted system folder paths that are already protected by restricted file and directory permissions.
M1047	Audit	Periodically audit virtual machines for abnormalities.
M1033	Limit Software	Restrict the installation of software that may be abused to create hidden desktops, such as hVNC, to user

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0502	Detection Strategy for Hidden Artifacts Across Platforms	AN1384	Abuse of file/registry attributes to hide malicious files, directories, or services. Defender view: detection of attrib.exe setting hidden/system flags, creation of Alternate Data Streams, or registry keys altering file visibility.
		AN1385	Hidden file creation using leading '.' or file attribute changes with chattr (immutable/hidden flags). Defender view: detect execution of chattr, lsattr anomalies, and unusual hidden files appearing in system directories.
		AN1386	Hidden files via 'chflags hidden' or Apple-specific attributes, LaunchAgents/LaunchDaemons placed in non-standard hidden directories. Defender view: detect command execution modifying file flags and unusual plist creation in hidden paths.
		AN1387	Abuse of VMFS or ESXi shell to hide datastore files, renaming/moving VMDK or VMX files into hidden directories. Defender view: anomalous ESXi shell commands or file operations obscuring VM artifacts.
		AN1388	Malicious macros or embedded objects hidden within Office documents by renaming streams or using hidden OLE objects. Defender view: detection of hidden macro streams or objects in documents correlated with anomalous execution.

References

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